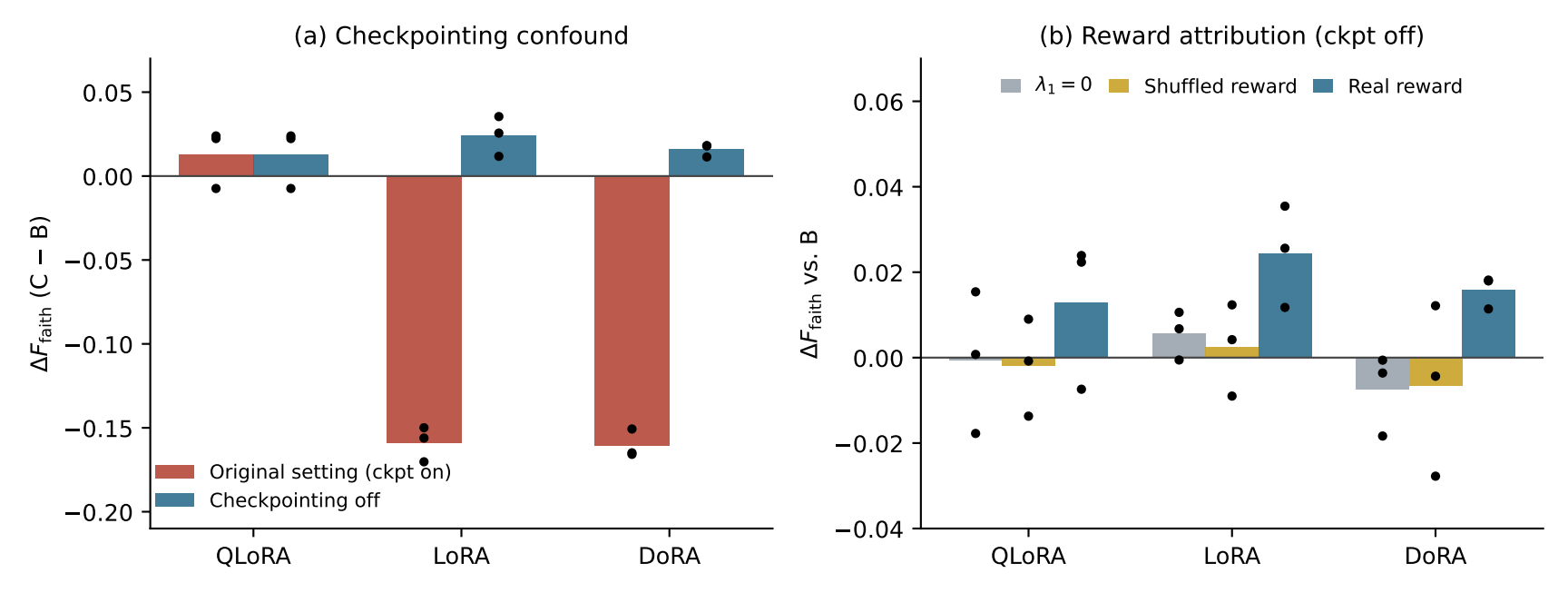
(a) Checkpointing confound
ΔF_{faith} (C − B)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
Original setting (ckpt on)
Checkpointing off
QLoRA
LoRA
DoRA
(b) Reward attribution (ckpt off)
$\lambda_1 = 0$
Shuffled reward
Real reward
ΔF_{faith} vs. B
0.06
0.04
0.02
0.00
−0.02
−0.04
QLoRA
LoRA
DoRA